Attribution ledger: a property belongs to level L only if it cannot be stated with the algebra of level L − 1

Already in $\mathbb{C}$

- amplitude–phase duality (90° rotation of the exponent plane)
- imaginary amplitude → indefinite norm, light cone
- two sections of the complex angle: $\cos^2$ (Born) and $\cosh\ \phi = r$
- self-dual point $|k| = 1$ ($\alpha = \pi/2$)
- $\Sigma z = 0$ and $\Sigma z^2 = 0$, $U^n = I$, counting m/n
- the division trap, linearity in the exponent
- amplitude ≠ curvature radius (visible on $S^1 \times \mathbb{R}$)

Added by $\mathbb{H}$

- amplitude and phase become distinguishable ($m \perp n$)
- non-commutativity $[m, n] = 2m \times n$ → third direction = space
- signature (1, 3) of the true square; light cone with direction
- two standing waves remember a position
- condensates become possible ($\mathbb{C}$ has no direction)
- 3-body discriminant $\Delta_3 = \frac{1}{4}|uv - vu|^2$

Added by $\mathbb{O}$

- non-associativity: associator needs 3; with closure, 4
- R-type and Q-type amplitudes become distinct → mass vs charge
- split form (4, 4) and $|a| = |b|$: the spacetime equation
- order of sequential bonding matters (history)
- no Dyson ensemble (no precedent)